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EE 133 Homework 3

1. (20pts) For $E - E_C = 0.01$ eV and $m_n^* = 1.18 m_0$ where m_0 is the mass of an electron, calculate the density of states $g_c(E)$ in units of $\left(\frac{1}{\text{eV}\cdot\text{cm}^3}\right)$. Hint: it is between 10^{20} and $10^{21} \left(\frac{1}{\text{eV}\cdot\text{cm}^3}\right)$.

2. When $E > E_F + 3k_B T$, we claim that we can approximate the Fermi factor

$$f(E) = \frac{1}{1 + e^{(E-E_F)/k_B T}}$$

with the Boltzmann factor

$$B(E) = e^{-(E-E_F)/k_B T}.$$

(a) (20pts) Derive an expression for the percent error $[B(E) - f(E)]/f(E)$ for *arbitrary* temperature > 0 .

Determine the actual percent errors for:

(b) (5pts) $E = E_F + 3k_B T$

(c) (5pts) $E = E_F + 4k_B T$

(d) (5pts) $E = E_F + 5k_B T$

Use the following values when needed for problems below.

Si: $E_G = 1.12$ eV

GaAs: $E_G = 1.42$ eV

Ge: $E_G = 0.66$ eV

$kT = 0.026$ eV at $T=300$ K

Use the following values for the density of states masses and bandgaps:

Material	DOS m_n^*/m_0	DOS m_p^*/m_0	E_G (eV)
Si	1.08	0.549	1.12
Ge	0.561	0.2915	0.66
GaAs	0.067	0.473	1.424

Other useful sources: Appendices II and III of Streetman & Banerjee. Tables and graphs in the Lecture notes.

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3. The condition $(E_v + 3kT) \leq E_F \leq (E_c - 3kT)$ separates two different regimes.
 - a. (5 pts) What are they?
 - b. (5 pts) Write down the 2 expressions for the electron density and the 2 expressions for the hole density valid when $(E_v + 3kT) \leq E_F \leq (E_c - 3kT)$.
4. (10 pts) **Calculate** n_i ($1/\text{cm}^3$) for intrinsic Si at $T=300\text{K}$. Use the density-of-states masses and bandgap in the table above.
5. (10 pts) **Calculate** n_i ($1/\text{cm}^3$) for intrinsic GaAs at $T=300\text{K}$. Use the density-of-states masses in the table above.
6. (a) (10 pts) (Determine the electron and hole density in Si when $(E_c - E_F) = 3 kT$ at $T=300\text{K}$.

(b) (10 pts) Determine the electron and hole density in Si when $(E_F - E_v) = 3 kT$ at $T=300\text{K}$.
7. At $T=300\text{K}$, a semiconductor has $E_G = 1.1 \text{ eV}$ and $N_C = N_V$. It is doped with $10^{15} / \text{cm}^3$ fully-ionized donors. E_F is 0.25 eV below E_C .
 - (a) (10pts) Draw the band diagram. Let $E_v = 0$. Label the energies of E_C , E_F , and E_i .
 - (b) (10 pts) Calculate n
 - (c) (10 pts) Calculate n_i
 - (d) (10 pts) Calculate p

For the next two problems, use the plot on slide 33 of the Wk3.pdf slides for the value of n_i .

8. A semiconductor device requires n-type material ($n \gg p$) at $T = 400\text{K}$.
 - (a) (10 pts) Would As doped Si at $10^{15} / \text{cm}^3$ meet the above criteria? Explain using equations.

(b) (10 pts) Would Sb doped Ge at $10^{15} / \text{cm}^3$ meet the above criteria? Explain using equations.
9. (10 pts) Si is doped with 10^{15} atoms / cm^3 of phosphorous. What are the electron and hole concentrations at $T=540\text{K}$?

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10. Determine the intrinsic Fermi level of GaAs at
(a) (10 pts) $T=300\text{K}$

(b) (10 pts) $T=400\text{K}$.